
Mass Reconstruction of Di-Tau Systems with Machine Learning

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September 2025

Abstract

Mass reconstruction of di-tau systems is a crucial task in searches for beyond Standard Model interactions. Among the factors limiting the sensitivity in these channels, is the presence of undetectable neutrinos in the final state, which analytical methods struggle to account for accurately. To address this, a machine learning approach was employed using both Deep Neural Networks and Transformers. The models were trained on kinematic and geometric features of the events, and an extended Transformer, incorporating interaction features between pairs of particles, was also considered to provide additional information. The analysis focused on two key channels: Vector Boson Fusion and Drell–Yan production of $Z \rightarrow \tau\tau$ events, with the Drell–Yan dataset enabling more robust model training due to its larger size. All machine learning models outperformed traditional methods, with Transformer architectures achieving the best performance in their respective channels. Nonetheless, the models exhibited inherent biases due to the peaked mass distribution and the presence of rare high-mass outliers. Feature importance provided clear insights into the decision-making of the Deep Neural Networks, whereas transformer-specific metrics, such as attention entropy and head diversity, offered limited interpretability.

Contents

1	Introduction	1
1.1	Hadronic Taus	1
1.2	Vector Boson Fusion and the Drell–Yan Process	2
1.3	Mass Reconstruction Methods	3
1.4	Monte Carlo Data	4
2	Machine learning	5
2.1	Dimensionality is Key	5
2.2	Deep Neural Networks	5
2.3	Transformers	7
2.4	Transformer with Interaction Tokens	9
3	Model metrics	9
3.1	Deep Neural Network Metrics	10
3.2	Transformer Metrics	10
4	Methodology	11
4.1	Training of Deep Neural Networks	12
4.2	Training of Transformers	12
5	Results and Analysis	13
5.1	VBF results	13
5.2	Drell–Yan results	16
5.3	VBF Models on Drell–Yan Data and Vice-Versa	19
6	Discussion	20
7	Conclusion	21
8	Appendix	24
A	Scaling and Distribution of Training Variables	24

1 Introduction

The Standard Model is currently the most accurate theory for describing particle interactions. Among its many successes, it predicts the electron’s dipole moment and the masses of the Z and W bosons with remarkable precision that agrees closely with experimental results [1]. However, as particle accelerators achieve higher energy scales and larger integrated luminosities, untested areas of the allowed phase space can be probed. The expansion of the phase space accessible to experiments has gone hand in hand with advancements in machine learning theory and hardware, which have enabled more precise analyses of complex dynamical systems and of the non-linear relationships between kinematic variables.

Despite these advances, some challenges persist. In particular, reconstructing events with tau leptons remains difficult because the neutrinos produced in the decay escape detection, leaving part of the energy unmeasured [2]. Methods for estimating the missing energy have been devised, such as the collinear approximation and the missing mass calculator, but both of these methods have significant drawbacks [3, 4]. The collinear approximation fails in certain event topologies, and the missing mass calculator performs better, but is computationally expensive.

This report aims to enhance the mass reconstruction of events in which a Z boson decays to two taus (di-tau) by applying machine learning techniques guided by physical principles.

Achieving higher sensitivity in this area would enable more precise studies of tau leptons, which, as the heaviest leptons, probe the most extreme regions of the phase space accessible at the LHC and play a crucial role in searches for beyond Standard Model particles such as Z' bosons (a heavier version of the Z boson) [5].

Lastly, this project is founded on the efforts of Hong Kai and Joseph Stevenson, alumni from the University of Manchester whose initial investigations of di-tau systems motivated the work presented here [6].

1.1 Hadronic Taus

When reconstructing di-tau systems from Standard Model processes, the two most relevant production channels are $Z \rightarrow \tau^+\tau^-$, with a cross section of 1.97 nb and, $H \rightarrow \tau^+\tau^-$, with a cross section of 55.5 pb, both at a centre-of-mass energy of $\sqrt{s} = 13$ TeV [7, 8]. Since the mass of the Z boson has been measured with higher precision than that of the Higgs boson, the $Z \rightarrow \tau^+\tau^-$ channel provides a more well-constrained benchmark for studies of di-tau reconstruction [9, 10]. The Z boson decays into di-tau systems with a branching ratio of approximately 3.4%, as illustrated in Figure 1a. Tau leptons themselves decay to hadronic final states about 65% of the time, and leptonically the remaining 35% of the time. In leptonic decays, the tau produces a charged lepton and two neutrinos, whereas in hadronic decays, the tau produces a neutrino accompanied by either one pion or three pions, as shown in Figure 1b.

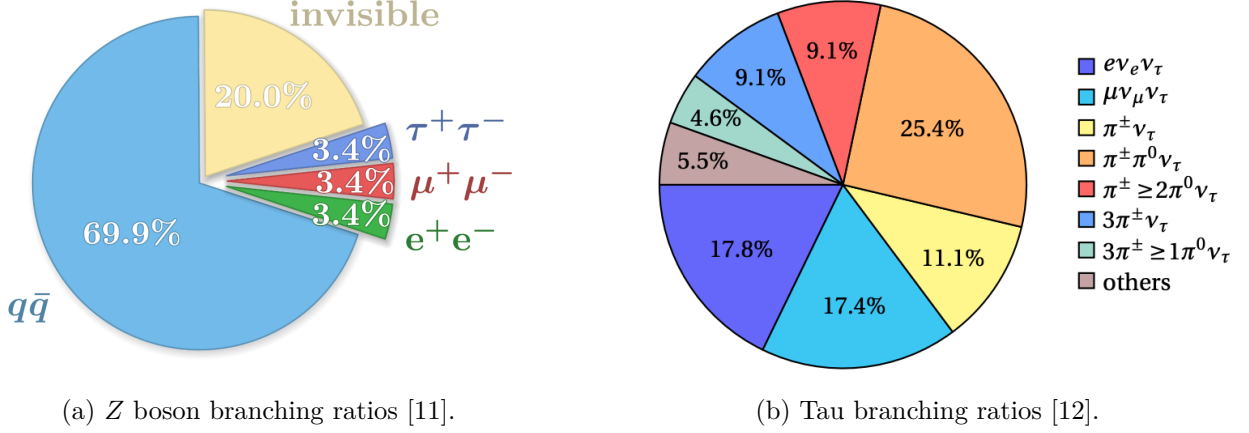


Figure 1: Decays of Z bosons and taus.

Electrons and muons from tau decays produce distinct signatures in the calorimeter, making them relatively easy to detect, although it is not always possible to determine unambiguously whether they originate from a tau. Furthermore, a significant fraction of the tau's energy is carried away by an additional neutrino, which escapes detection. In contrast, taus decaying to hadronic final states are more challenging to tag because of the large QCD background. Nevertheless, they can be identified by their characteristic one- or three-prong decay topologies [13].

The channel $Z \rightarrow \tau_\ell \tau_h$ is particularly well-suited for this study, as it offers a compromise between the number of undetectable neutrinos and the ease of tagging leptons, while still maintaining a manageable QCD background from the hadronic tau. This balance allows for sufficient event statistics, whereas the $Z \rightarrow \tau_h \tau_h$ channel becomes inaccessible due to the much higher trigger thresholds required [14].

1.2 Vector Boson Fusion and the Drell–Yan Process

Two key production mechanisms in the $Z \rightarrow \tau_\ell \tau_h$ channel are Vector Boson Fusion (VBF) and the Drell–Yan process.

Although the VBF process has a relatively low cross section, it is particularly valuable due to its distinctive topology, which enables precise event calibration. In VBF events, as illustrated in Figure 2a, two incoming quarks emit force carriers such as W bosons, which then fuse to produce a Z or H boson that subsequently decays. This t -channel process is characterized by a large pseudorapidity gap $\Delta\eta$, as the quarks are only slightly deflected from their original trajectories. Consequently, the central region of the detector along the beamline exhibits minimal activity, isolating the decay products of the Z boson from the forward jets. The two forward jets typically have high transverse momentum (p_T), and combined with the relatively empty central region, this results in a distinctive experimental signature with low background contamination [15].

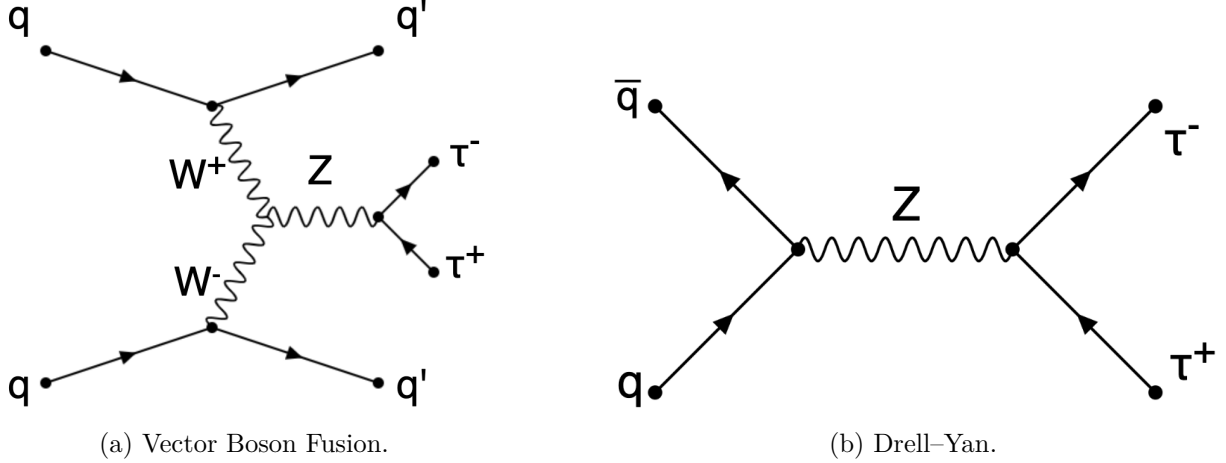


Figure 2: Feynman diagrams for relevant $Z \rightarrow \tau^+\tau^-$ processes.

In contrast, the Drell–Yan process illustrated in Figure 2b, has a significantly larger cross section than the VBF and therefore dominates the $Z \rightarrow \tau^+\tau^-$ production rate at the LHC. In this process, a quark and an antiquark annihilate via an intermediate Z boson, which subsequently decays into a lepton pair. Although the underlying hard scattering is electroweak, the process is accompanied by substantial QCD activity. Initial-state gluon radiation and subsequent parton showers produce additional jets in the event, complicating the reconstruction of the $\tau^+\tau^-$ invariant mass [16].

An additional challenge arises from reducible backgrounds, where hadronic jets are misidentified as hadronic tau decays (“jet fakes”). These backgrounds are particularly significant in the Drell–Yan channel, as the dense hadronic environment increases the likelihood of misidentification.

1.3 Mass Reconstruction Methods

The mass of the Z boson system can be estimated using the kinematic information of the identified decay products. However, this reconstruction is incomplete due to the presence of undetected neutrinos, which carry away a portion of the energy. A commonly used approach to approximate the neutrino momenta is the collinear approximation, which assumes that the neutrinos are emitted in the same direction as the visible decay products. Under this assumption, the transverse momenta of the neutrinos can be solved using Equation 1, where E_T^{miss} denotes the missing transverse momentum (MET) as illustrated in Figure 3 [3].

$$p_{\nu_l} + p_{\nu_h} = E_{\text{miss}}^T \quad (1)$$

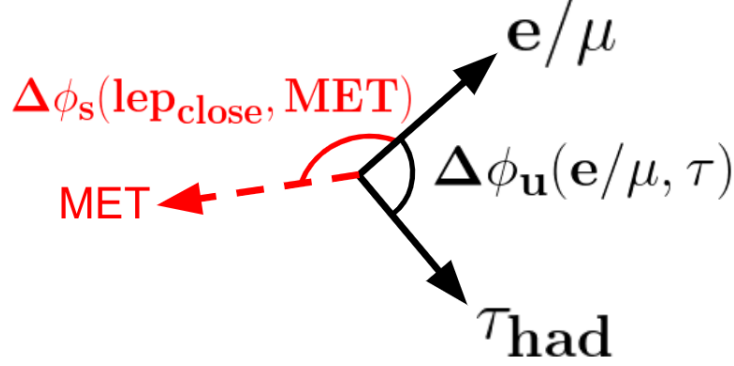


Figure 3: Event kinematics of Z decay to a hadronic tau and a lepton. The angles are labelled signed (s) and unsigned (u) respectively. $\Delta\phi_s(\text{lep}_{\text{close}}, \text{MET})$ is positive when the MET is inside the event, and negative when it is outside as shown.

Finally, the mass of the di-tau system can be approximated using Equation 2. Notably, a correction of 5 GeV has been shown to improve the accuracy of the mass reconstruction in cases where the MET lies outside the opening angle defined by the two visible decay products [17]:

$$M_{Z\text{-reco}} = \begin{cases} \sqrt{(p_{\text{had}} + p_{\text{lep}} + \nu_{\text{lep}} + \nu_{\text{had}})^2}, & \text{if MET inside,} \\ \sqrt{(p_{\text{had}} + p_{\text{lep}} + \nu)^2} + 5 \text{ GeV}, & \text{if MET outside.} \end{cases} \quad (2)$$

p and ν represent the four-vectors of the leptons and neutrinos respectively. Another method, known as the Missing Mass Calculator (MMC), seeks to improve upon the assumptions of the collinear approximation by probabilistically estimating the mass of the di-tau system. The MMC operates by scanning over the kinematically allowed phase space of the undetected neutrinos and assigning likelihoods to different momentum configurations based on the decay kinematics and detector resolution. By integrating over these probabilities, the method produces the most likely estimate of the di-tau invariant mass [4].

Throughout this work, masses reconstructed with the collinear approximation are denoted as $M_{Z\text{-reco}}$ and, ones obtained from the Missing Mass Calculator are denoted $M_{Z\text{-mmc}}$.

1.4 Monte Carlo Data

Monte Carlo (MC) simulation data is invaluable in high-energy physics analyses as it provides a fast and accurate way to simulate rare events with high fidelity to the ATLAS detector. In addition, MC data provides access to generator-level variables that describe the true event kinematics before detector effects are applied. One such variable is the true mass of the Z boson prior to its decay, which is not available in real experimental data. This allows us to use the true mass, denoted as $M_{Z\text{-true}}$, to train an algorithm that can subsequently be applied to real ATLAS run data.

The MC event data is subjected to a series of selection cuts to isolate high-quality $Z \rightarrow \tau_{\text{had}} \tau_{\text{lep}}$

events and reduce backgrounds from misidentified objects. First, only leptons passing isolation and identification criteria are selected, and events consistent with $\tau_{had} e$ or $\tau_{had} \mu$ topologies are retained. Events containing any b-jets are rejected to suppress the $t\bar{t} \rightarrow WbWb$ background, which produces similar final states but with additional b-quarks [18]. Events are further required to contain at least two light jets to preferentially select geometries in which the two taus are not back-to-back. Finally, an angular selection is applied to remove events in which the missing transverse momentum (MET) forms an angle greater than 90° relative to the nearest lepton; for example, the event depicted in Figure 3 would not pass this cut. This requirement ensures physically reasonable event configurations.

After applying these cuts, the resulting dataset contains 194,430 events for the VBF samples and 6,225,580 events for the Drell–Yan samples.

2 Machine learning

The field of machine learning (ML) has rapidly transformed both everyday life and the way scientific research is conducted. Thanks to advances in hardware, algorithms originally conceived in 1943, which were previously too computationally expensive, can now be leveraged to achieve improvements in accuracy and speed compared to traditional algorithms [19].

Particle physics has similarly embraced this trend, as evidenced by the increasing adoption of ML algorithms in Run 3 of ATLAS, replacing many of the analytical methods used in Runs 1 and 2. One example important to this analysis is the recurrent neural network employed to discriminate genuine tau signals from fake tau candidates by combining tracking information with calorimeter data [20].

2.1 Dimensionality is Key

In general, given appropriate data and training, ML approaches have the potential to outperform analytical algorithms due to their ability to process high-dimensional data and learn non-linear relationships between subsets of parameters that may be difficult to identify manually [21].

This capacity to simplify and distil important connections in increasingly complex datasets, as the number of model parameters grows, is what makes ML particularly well-suited to problems in particle physics. In event-level ML simulations, a small number of initial states and observables evolve through complex quantum-mechanical processes, producing intermediate states that are orders of magnitude more numerous. These intermediate states then pass through simulated detectors, generating high-dimensional outputs in which the correlations between the initial and final states are subtle and often challenging to extract by conventional means.

2.2 Deep Neural Networks

A deep neural network (DNN) consists of a sequence of layers, namely the input layer, one or more hidden layers, and the output layer, as illustrated in Figure 4b. These layers are composed

of smaller building blocks called perceptrons. In a DNN, each perceptron is fully connected to all perceptrons in the subsequent layer, and each connection includes an associated bias term.

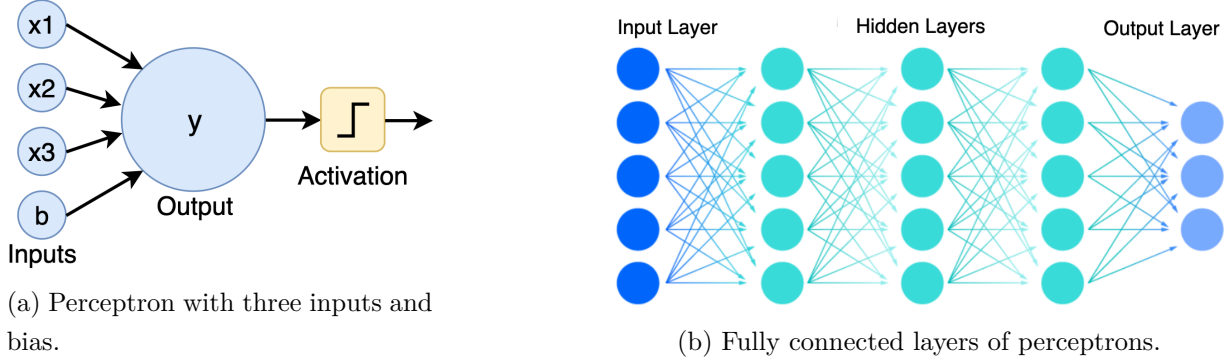


Figure 4: Architecture of a Deep Neural Network [22].

As shown in Figure 4a and expressed in Equation 3, an artificial neuron receives a set of input values $x_i, \dots, x_n$, computes a weighted sum with the corresponding connection weights ω_{ji} , and adds a bias term b_j . Here, the index i denotes the i -th input to the j -th perceptron in the layer [23].

$$y_j^{(l)} = f\left(\sum_{i=1}^n w_{ji}^{(l)} x_i^{(l)} + b_j^{(l)}\right) \quad (3)$$

The weighted sum computed by a perceptron is then passed through a non-linear activation function f , producing the output of perceptron j , denoted y_j . Importantly, the activation function is what distinguishes a DNN from a purely linear transformation, preventing it from being equivalent to a high-dimensional linear regression problem.

In a DNN, the inputs are processed layer by layer through the mechanism described above. Specifically, the outputs of the l -th layer, $y_j^{(l)}$, serve as the inputs to the $(l+1)$ -th layer. This procedure, known as forward propagation, is repeated until the output layer is reached, yielding the network's predictions.

The performance of the model is quantified by comparing these predictions to the true data using a loss function, $\mathcal{L}$ (e.g. mean squared error). During training, the model parameters (weights and biases) are iteratively updated to minimize the loss function, as expressed mathematically in Equations 4 and 5 [24]. This procedure, known as backpropagation, relies on gradients of the loss function with respect to the model parameters to “descend” towards better performance. The updates to the weights $\omega_{ji}^{(l)}$ and biases $b_j^{(l)}$ are scaled by the learning rate α , and this process is repeated over multiple epochs until a satisfactory loss value is achieved.

$$\frac{\partial \mathcal{L}}{\partial w_{ji}^{(l)}} = \frac{\partial \mathcal{L}}{\partial y_k^{(L)}} \cdot \frac{\partial y_k^{(L)}}{\partial y_{k'}^{(L-1)}} \cdots \frac{\partial y_j^{(l)}}{\partial w_{ji}^{(l)}} \quad (4)$$

$$w_{ji}^{(l)} \leftarrow w_{ji}^{(l)} - \alpha \frac{\partial \mathcal{L}}{\partial w_{ji}^{(l)}}, \quad b_j^{(l)} \leftarrow b_j^{(l)} - \alpha \frac{\partial \mathcal{L}}{\partial b_j^{(l)}} \quad (5)$$

The updates to the weights and biases can be optimized using more sophisticated strategies than simple gradient descent. These methods allow the training to converge more quickly in regions where the loss function behaves monotonically and help avoid getting trapped in local minima when the loss landscape is complex. Such strategies are implemented through optimizers, with adaptive moment estimation (ADAM) being one of the most widely used.

It is also important to note that as the model trains, its performance is monitored after each epoch using a validation dataset. Once training is complete, the final performance is evaluated on an independent test dataset consisting of previously unseen events.

2.3 Transformers

Since the early days of machine learning, tackling problems in which the data is sequential has been a challenging task. Recurrent neural networks (RNNs) were among the first to tackle this by maintaining a hidden state across adjacent steps, but they often struggled to learn long-range dependencies [25]. Long Short-Term Memory networks (LSTMs) addressed this limitation with memory cells and gating mechanisms that help retain information over longer sequences, though training remained slow and capturing very distant dependencies was still challenging [26]. The introduction of transformers marked a major breakthrough: by using self-attention, they can process sequences in parallel and model relationships between all elements, achieving remarkable results in fields like computer vision and natural language processing with models like ChatGPT [27].

One of the key theoretical results that underpins this success, is a theorem from linear algebra stating that as the dimensionality of a space increases, the number of nearly orthogonal vectors grows exponentially [28]. Conceptually, if each orthogonal vector in the input space is thought of as encoding a distinct unit of meaning, then nearly orthogonal vectors provide close approximations to such units. This exponential growth in representational capacity enables transformers to approximate highly complex systems.

The transformer architecture employed in this study is illustrated in Figure 5 and follows an encoder-only configuration. In this setup, the input data is first divided into tokens, with each token corresponding to a single feature (e.g. energy E , pseudorapidity η , azimuthal angle ϕ , etc.). Each token is then passed through an input embedding network, implemented as a deep neural network, which maps the low-dimensional features into a higher-dimensional space. The resulting high-dimensional token vectors are concatenated and fed into the transformer block [27]. Although the transformer block preserves the overall dimensionality of the input matrix, it updates the values of the token vectors, thereby enhancing their representational power. Finally, a pooling layer aggregates these vectors into a single vector, which is then provided as input to a DNN responsible for the regression task.

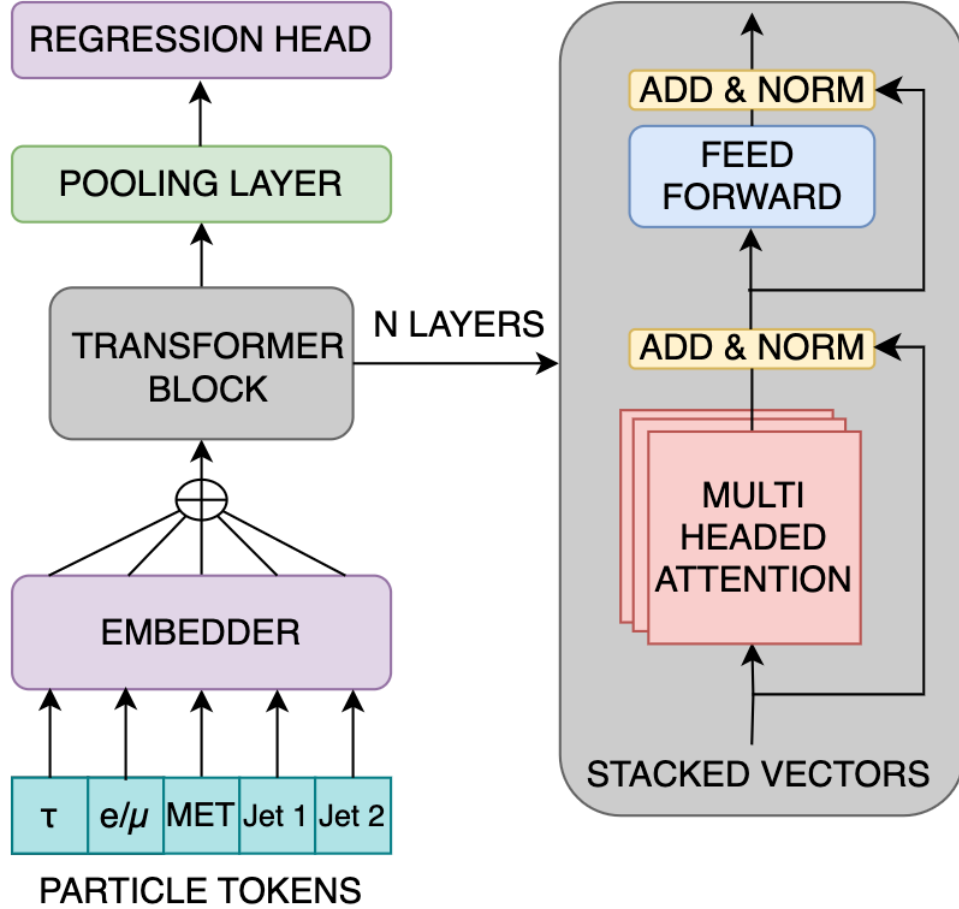


Figure 5: Architecture of an encoder-only transformer. The transformer block is broken down into its constituents parts that form a layer. In purple are blocks in which DNNs are used.

The transformer block is built around the self-attention mechanism, which evaluates the similarity between high-dimensional token vectors in the context of a given task. Geometrically, this operation can be viewed as shifting and rotating the token vectors within the high-dimensional feature space to enhance their predictive power before they are passed to the pooling layer. Returning to the earlier analogy of each axis encoding a unit of meaning, the transformer block refines the “meaning” assigned by the embedding layer by considering each token both individually and in relation to all other tokens. In natural language processing, this corresponds to contextualizing words based on their neighbours; in particle physics, while there is no direct interpretation, the mechanism serves to enrich the event representation by incorporating correlations among the kinematics of all reconstructed objects.

Self-attention is implemented using multiple attention heads. Each head projects the high-dimensional token vectors into a lower-dimensional subspace, enabling the model to capture simple yet complementary patterns. Different heads learn distinct relationships which are then concatenated back into the original feature space, forming the multi-head attention mechanism. The transformer block is composed of repeated layers of this mechanism, combined with residual

connections, layer normalization, and position-wise feed-forward networks. A detailed description of these components and their computations can be found in [27].

2.4 Transformer with Interaction Tokens

Previous studies have demonstrated that incorporating pairwise particle interaction features can enhance the performance of jet-tagging algorithms [29]. These features are defined as shown in Equation 6, where a and b denote two distinct particles in the pair, y is the rapidity and $\vec{p}$ is the three-momentum.

$$\begin{aligned}\Delta &= \sqrt{(y_a - y_b)^2 + (\phi_a - \phi_b)^2}, \\ k_T &= \min(p_{T,a}, p_{T,b}) \Delta, \\ z &= \frac{\min(p_{T,a}, p_{T,b})}{p_{T,a} + p_{T,b}}, \\ m^2 &= (E_a + E_b)^2 - \|\vec{p}_a + \vec{p}_b\|^2.\end{aligned}\tag{6}$$

To incorporate these features, a separate embedding is constructed, since they differ from the individual particle features. After being mapped into the same high-dimensional space as the particle tokens, the resulting vectors are concatenated with the particle embeddings and collectively passed to the transformer block.

3 Model metrics

As shown in Figure 6, the data distribution is sharply peaked around the Z boson mass but also exhibits long tails, with events extending to energies several orders of magnitude larger than the Z peak. For this reason, the median absolute error (MAE) and the coefficient of determination (R^2) were chosen as evaluation metrics. These metrics provide both robustness to large outliers and sensitivity to the strong concentration of events near the peak. Their definitions are given in Equations 7 and 8 [30]:

$$\text{MAE} = \text{median}_i (|y_i - \hat{y}_i|),\tag{7}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.\tag{8}$$

The performance of M_Z -reco and M_Z -mmc on the VBF test dataset is shown in Figure 6. Interestingly, both methods achieve comparable peak resolution. However, the collinear approximation performs poorly in terms of the R^2 metric, yielding a negative score. This indicates that its predictions are worse than simply using the mean, primarily due to the extreme outliers it produces when the neutrinos are emitted nearly back-to-back.

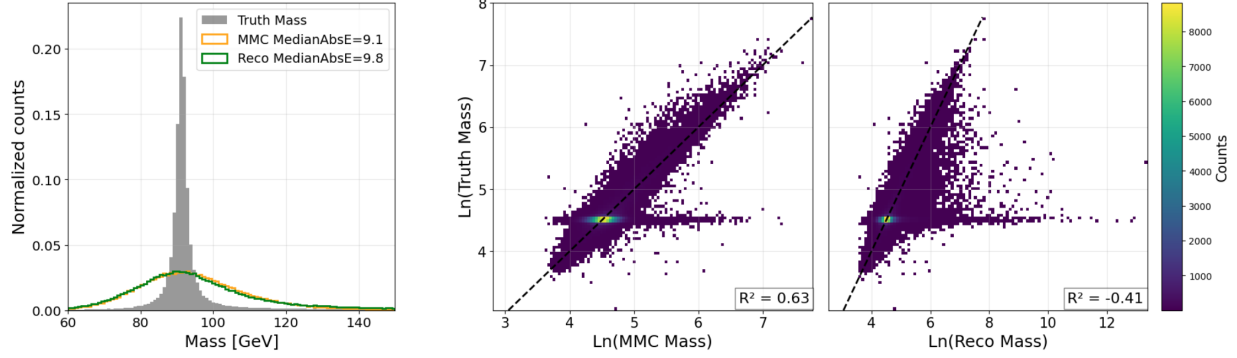


Figure 6: Performance of M_Z -reco and M_Z -mmc on the testing dataset. A black dashed line indicates perfect correlation between the prediction and truth mass.

3.1 Deep Neural Network Metrics

Feature importance in deep neural networks (DNNs) quantifies the contribution of each input variable to the model’s predictive performance. Intuitively, it indicates which features the model relies on most when making predictions. A common approach for estimating feature importance is permutation importance. In this method, the values of a single input feature are randomly permuted, which breaks their correlation with the target, while all other features remain unchanged [31]. If the model’s predictive error increases substantially after this perturbation, the permuted feature is deemed important for accurate prediction. Conversely, features whose permutation produces little to no change in error are less relevant.

3.2 Transformer Metrics

To better understand the decisions made by a transformer, it is instructive to examine its core component: the attention mechanism. By averaging the attention weights across the heads in the multi-head attention block, one can quantify the attention entropy and diversity within each layer.

Consider the vector of attention weights for head h in layer l , denoted as $\mathbf{A}^{(l,h)} = [a_1, a_2, \dots, a_n]$, where there are n tokens to attend to and the vector is normalized such that $\sum_{i=1}^n a_i = 1$. The entropy for a single head is then defined as:

$$S^{(h,l)} = -\sum_{i=1}^n a_i \log a_i, \quad (9)$$

which can subsequently be averaged across all heads in the layer to characterize the overall attention distribution [32]. Entropy quantifies how concentrated the attention of a head is. Lower entropy indicates that the model focuses on fewer tokens, implying higher confidence, whereas higher entropy reflects a more diffuse attention distribution, where the model considers a larger number of tokens. While diffuse attention can lead to richer representations, a model that assigns nearly equal attention to all tokens may not have learned meaningful patterns. The entropy

ranges from 0 to $\ln n$, with high values corresponding to attention spread across all tokens.

Attention diversity, on the other hand, measures the degree of difference between the attention weights of different heads within the same layer. It is computed using cosine similarity between head pairs, where H is the number of heads in the layer:

$$D^{(l)} = \frac{2}{H(H-1)} \sum_{h_1 < h_2} \left[1 - \cos(\angle(\mathbf{A}^{(l,h_1)}, \mathbf{A}^{(l,h_2)})) \right]. \quad (10)$$

The diversity ranges from 0 to 1. Low diversity indicates that heads attend to similar tokens in similar ways, potentially suggesting redundancy, while high diversity indicates that heads focus on different aspects of the input [33]. Returning to the geometrical interpretation of attention, high diversity implies that each head focuses on a different subspace of the high-dimensional embedding, resulting in a richer overall representation when the outputs of the attention heads are concatenated.

It is important to clarify that the attention weights themselves are not directly learned during training. Instead, three matrices—the key, value, and query matrices—are learned. The attention weights are dynamically computed as a function of these matrices and the embedded tokens, and therefore depend on the specific input data.

4 Methodology

When training both the DNN and transformer models, a simple architecture was initially employed. If the model underfit the data, its complexity was gradually increased. For the DNN, this involved enlarging the layer sizes and the input embedder. For the transformer, this included increasing the input embedder output dimension, the number of attention heads, and the number of transformer layers. Conversely, if the model overfit the data or exhibited signs of guessing such as a pile-up of events at the Z boson mass, indicating blind predictions, regularization techniques such as early stopping, layer and batch normalization, and increased dropout were used. New features and particle types, such as jets, were also introduced sequentially, retaining only those that improved the model metrics.

The models presented in this study represent a subset of those that successfully fit the data. Notably, even models with suboptimal architectures or hyperparameter choices can often fit the data effectively [34]. The models shown here were selected for both their performance and ease of training.

Training was performed using 80% of the data, with 10% reserved for validation and the remaining 10% for testing. Methods to subsample the dataset to prioritize specific event categories, such as those in the tails, were also explored (e.g., inverse KDE sampling), but they did not yield significant improvements relative to the added complexity [35].

Overall, the training time for the models listed in Table 1 primarily depended on the size of the data samples. For the VBF events (195k examples), training times ranged from 2 to 10 minutes for both DNN and transformer models, with larger parameter counts requiring more time. For the Drell–Yan dataset (6.2M examples), training times increased to approximately 20 minutes to 2 hours.

Model	Particle Level Variables	Event Level Variables	Num. Parameters
VBF DNN	E, p_T, η	$M_Z\text{-reco}, \Delta\phi_s(\text{lep}_{\text{close}}, \text{MET}), \Delta\phi_u(e/\mu, \tau)$	19 K
VBF Transformer	$E, p_T, \eta, \phi, q, \text{type}, \text{btag}$	$M_Z\text{-mmc}$	1199 K
VBF Inter-Transformer	$E, p_T, \eta, \phi, q, \text{type}, \text{btag}$	Pair-Interactions	1207 K
Drell–Yan DNN	E, p_T, η	$M_Z\text{-reco}, \Delta\phi_s(\text{lep}_{\text{close}}, \text{MET}), \Delta\phi_u(e/\mu, \tau)$	19 K
Drell–Yan Transformer	$E, p_T, \eta, \phi, q, \text{type}, \text{btag}$	None	1199 K
Drell–Yan Inter-Transformer	$E, p_T, \eta, \phi, q, \text{type}, \text{btag}$	Pair-Interactions	1207 K

Table 1: Overview of variables used as inputs for different models. The particle level variables are namely energy (E), transverse momentum (p_T), pseudorapidity (η), angle (ϕ), charge (q), type of particle and btag which indicates the type of jet if the particle is a jet. In addition to the collinear and MMC mass, at the event level there are the opening angles as seen in Figure 3 and the pair interactions described in Equation 6.

4.1 Training of Deep Neural Networks

Fewer, higher-level features were found to be most effective. As shown in Table 1, the particle-level ϕ variable was removed in favour of event-level opening angle parameters. For DNN models, the particle type variable was also omitted. This is because DNNs process all inputs simultaneously and, due to their architecture, implicitly recognize which features correspond to which particle (e.g., the first 5 features belong to one particle, while features 5–10 belong to another). It was also observed that feeding highly correlated or nearly duplicate features can hinder the model’s ability to learn meaningful patterns in relation to other variables. For instance, the most successful models included either $M_Z\text{-reco}$ or $M_Z\text{-mmc}$ as additional inputs, but not both.

Regarding architecture, wider networks were found to outperform deeper ones. In general, the most successful models employed two hidden layers with 128 perceptrons each. Interestingly, scaling the target variable was not beneficial. The models still achieved strong performance while regressing masses spanning 40–1400 GeV, covering three orders of magnitude.

4.2 Training of Transformers

Unlike the DNN, a more raw-data approach was effective for the transformer model. Raw variables such as btag, charge, and particle type were included. Notably, only jets contain btag information; for electrons, taus, and MET, placeholder values were used, which the model learned to ignore during training. For the type variables, it was important that the input embedder distinguished tokens corresponding to different particles, including the two jets.

In contrast to the DNN, scaling the target variable was essential for the transformer model, and

standard scaling (StandardScaler) was applied; see Appendix A for a mathematical description of this transformation. The input embedder generally consisted of two hidden layers with 64 and 128 perceptrons, 8 attention heads, and 2 transformer layers, providing a good balance between model complexity, number of parameters, and performance. The resulting embedded high-dimensional token vectors had dimension 128. The pooling layer employed mean averaging, and the output head contained two hidden layers of sizes 128 and 32, followed by a single regression perceptron as the output layer. When incorporating additional features, such as M_Z -reco, it was found more effective to append them to each token (duplicating the value) rather than creating a separate smaller embedder for these event-level features.

It was observed that training the transformer on the VBF dataset was more unstable, whereas a larger fraction of transformer models converged successfully on the much larger Drell–Yan dataset. Higher dropout rates (0.20–0.30) could be applied effectively for the Drell–Yan data. For the interaction-token transformer, the separate embedder for interaction features used the same structure with two hidden layers with 64 and 128 neurons, with no additional attention heads or layers. With 5 particle inputs, this resulted in a total of 10 pairwise interaction tokens.

5 Results and Analysis

5.1 VBF results

The performance of the three VBF models listed in Table 1 is shown in Figure 7. Notably, all three models demonstrate a significant improvement over analytical methods such as the collinear approximation and the missing mass calculator. Among the models, the DNN and the standard transformer fit the shape of the Z -peak most accurately, with the DNN achieving the lowest median absolute error of 2.47 GeV. The interaction transformer, by contrast, slightly underfits the central peak while overfitting the tail regions.

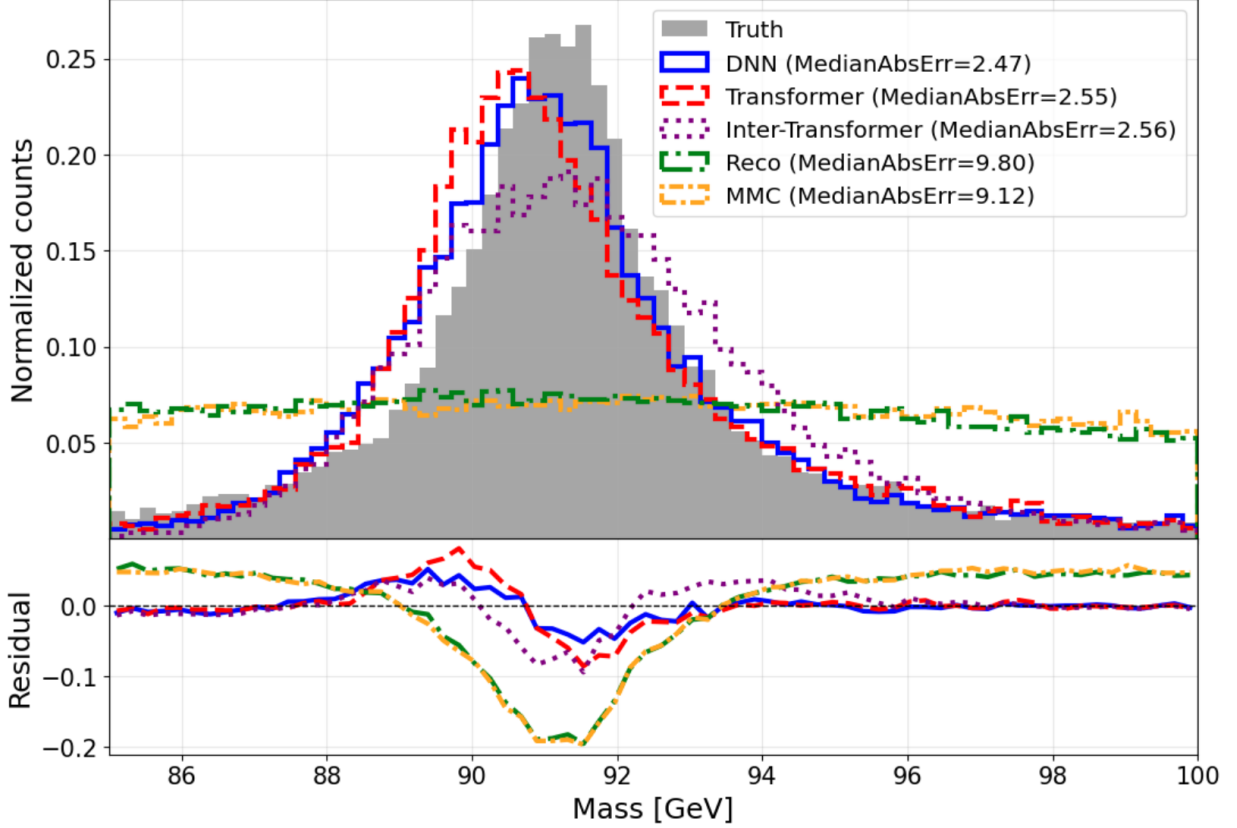


Figure 7: Performance of three ML models on VBF test data in the region 85–100 GeV. The results from the collinear approximation and missing mass calculator are also plotted.

Overall, all three models struggle with reconstructing the low-mass left-tail events, as seen in Figure 8. The bottom-left region is under-represented in the DNN and interaction transformer predictions, while the standard transformer achieves modest resolution. In contrast, high-mass right-tail events are reconstructed more reliably. However, for the highest-mass events (650 GeV and above), all models display vertical features, indicating that the model predicts the same value regardless of the true mass. This behavior arises because these events are rare in the training set, so the model cannot reliably learn their characteristics and instead focuses on the dense peak region.

More concerning are horizontal and vertical features around the Z -peak. Horizontal features indicate that the model predicts highly variable masses for events with similar true mass, while vertical features suggest that the model simply defaults to predicting the peak value to minimize the loss function, rather than accurately reconstructing the mass. Residual analysis (not shown) confirms these patterns.

Considering the combination of median absolute error (2.55 GeV), the highest R^2 score of 0.81, the weakest vertical stripe artifacts near the Z -peak, and the best low-tail resolution, the standard transformer is identified as the best-performing model for the VBF dataset.

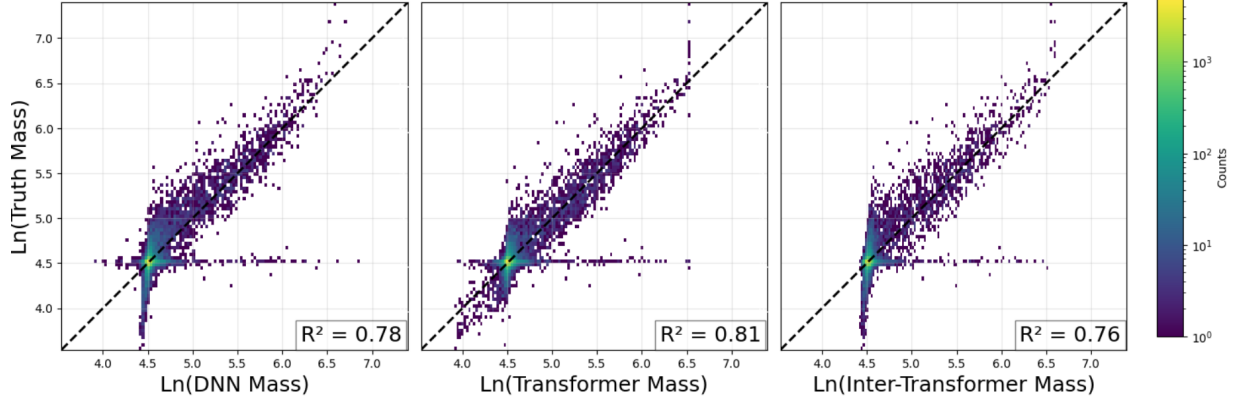


Figure 8: Logarithmic 2D correlation plots of VBF truth test data versus predictions from each of the three ML models. Plots share identical axes and colour maps.

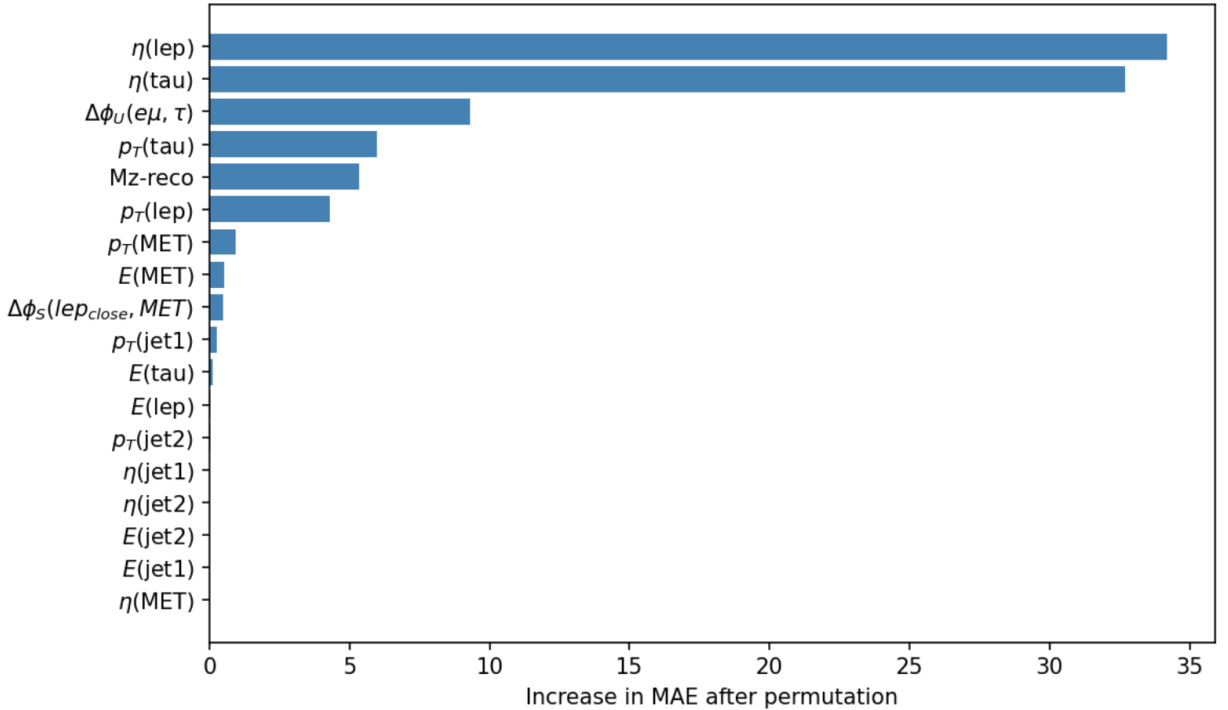


Figure 9: Feature importances for the DNN on VBF test data. The absolute increase in median absolute error is measured relative to the baseline of 2.47 GeV (Figure 7).

Although not the top-performing model, analysing the DNN remains instructive. The feature importance plot in Figure 9 shows that particle-level features dominate, while the energy, η , and p_T of the two jets contribute less to the prediction. Interestingly, the energies of the tau and lepton are also relatively unimportant, likely because the model can infer these values from other correlated observables, such as the η and p_T of the respective particles. The collinear mass is also weighted significantly, despite its poor performance in certain events (Figure 6). Opening angles are likewise important, as they define event kinematics and can aid in reconstructing transverse

momenta or energies.

A feature indicating whether the MET is inside or outside was also tested, but the model assigned it negligible importance. Consequently, it was removed from the final input set.

Analysing the behavior of the transformer models is more challenging, but attention entropy and diversity metrics (Table 2) provide insight. Both transformers exhibit high entropy in layer 0, close to the maximum possible value (1.6 and 2.7 respectively), which decreases slightly in layer 1. This positively indicates that the model initially distributes attention broadly across tokens before compressing information in subsequent layers. However, diffused attention can also suggest that the model does not differentiate effectively between embedded tokens.

For the basic transformer, layer 0 has near-zero diversity. Coupled with high entropy, this implies the first layer is effectively ‘frozen’ and does not contextualize the tokens. While this might appear suboptimal, previous work has shown that such frozen first layers are not uncommon and can act as filters, allowing later layers to more effectively contextualize the information [36]. This freezing often arises due to diminishing gradients during backpropagation, and it can be mitigated using schemes that boost the learning rate for initial layers.

Finally, both transformers exhibit good diversity in their second layer, indicating that these layers effectively learn distinct patterns across attention heads.

Metric	Transformer	Inter-Transformer
Ln(Num. Input Tokens)	1.6	2.7
Layer 0 Entropy	1.6	2.4
Layer 1 Entropy	1.5	1.9
Layer 0 Diversity	0.056%	28%
Layer 1 Diversity	18%	54%

Table 2: Entropy and Diversity metrics for each of the layers in the basic and interaction transformer trained on VBF data. The entropy is bounded between 0 and the natural log of the number of input tokens.

5.2 Drell–Yan results

Unlike the VBF dataset, the Drell–Yan sample contains approximately 6.2 million events, providing ample opportunity for the models to learn. This is reflected in the visibly improved performance shown in Figure 10. The DNN achieves the lowest median absolute error of 2.10 GeV, although it slightly overestimates the Z -peak. As with the VBF data, the models struggle more with the low-mass left tail than the right tail, as indicated by the large residuals in the 86–88 GeV region.

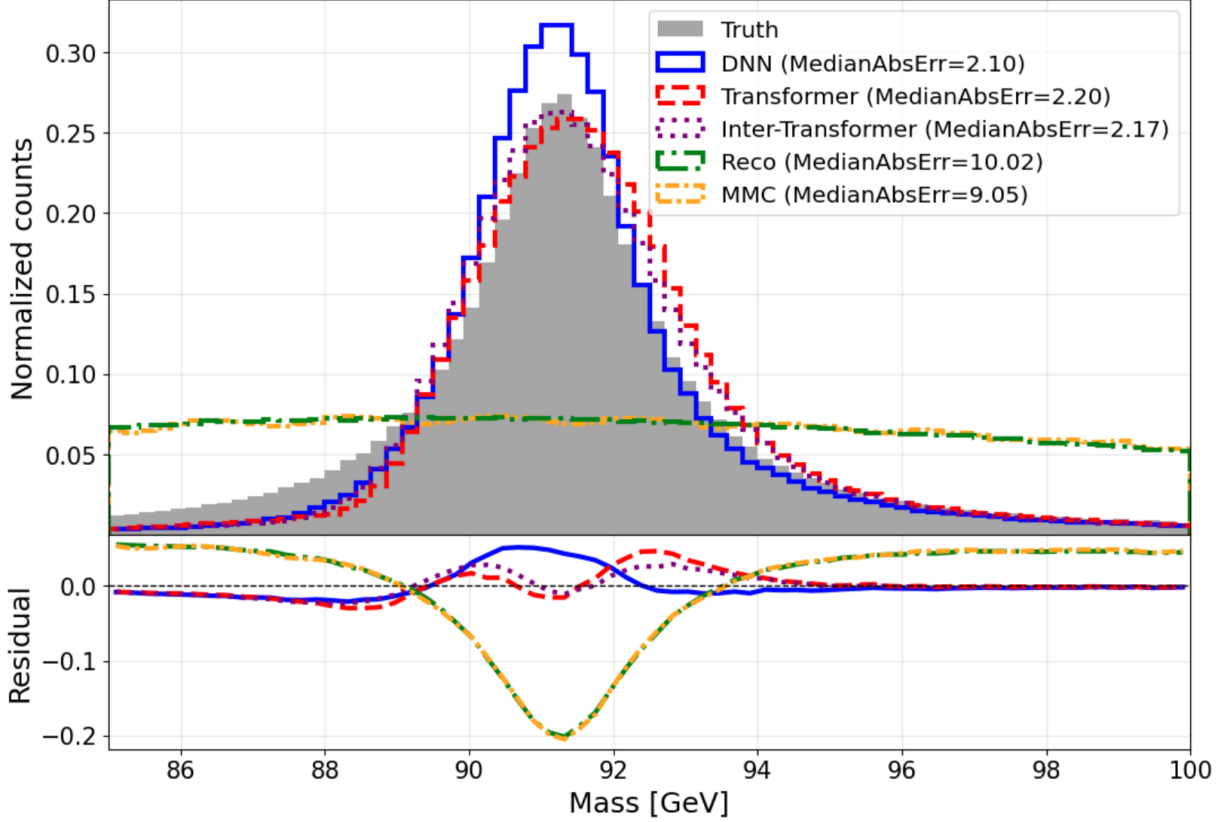


Figure 10: Performance of the three ML models on Drell–Yan test data in the region 85–100 GeV. The results from the collinear approximation and missing mass calculator are also plotted.

Figure 11 shows that all three models can reconstruct parts of the left tail but struggle with the very low-mass region (20 GeV) and the high-mass region (>800 GeV), due to the sparsity of these events in the training set. Both the interaction and basic transformers achieve an R^2 score of 0.77, indicating that the models explain approximately 77% of the variance in the data. Horizontal features are more prominent than in the VBF case, a consequence of the $30\times$ larger dataset, while vertical features around the Z -peak are subtler. The DNN slightly overfits the peak, but this is difficult to distinguish from the background, making it unclear whether the model is merely guessing the Z mass, as observed for VBF, or compacting events near the peak.

The interaction transformer generally performs better than the basic transformer, although it is difficult to select a single best model between it and the DNN. With the larger dataset, the models clearly avoid predicting low mass for high-mass events and vice versa, as indicated by the empty diagonal quadrants around the central Z -peak. Interestingly, despite the improved visual performance and lower median absolute error compared to VBF, the R^2 scores are slightly worse. This is expected, as the larger dataset increases the likelihood of rare, extreme outliers, which penalize the quadratic R^2 metric.

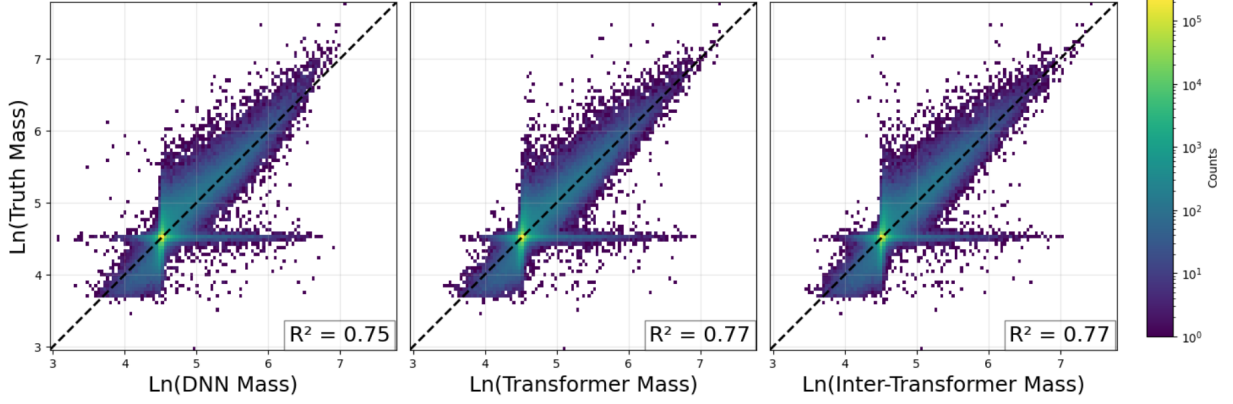


Figure 11: Logarithmic 2D correlation plots of Drell–Yan truth test data versus predictions from each of the three ML models. Plots share identical axes and colour maps.

Feature importance for the DNN is shown in Figure 12. The most important features are largely consistent with the VBF case (Figure 9), although their relative importance is slightly higher, while the η of the lepton and tau is slightly less important. As before, the model can infer certain kinematic variables from correlated features, since the energies of the tau and lepton are assigned low importance.

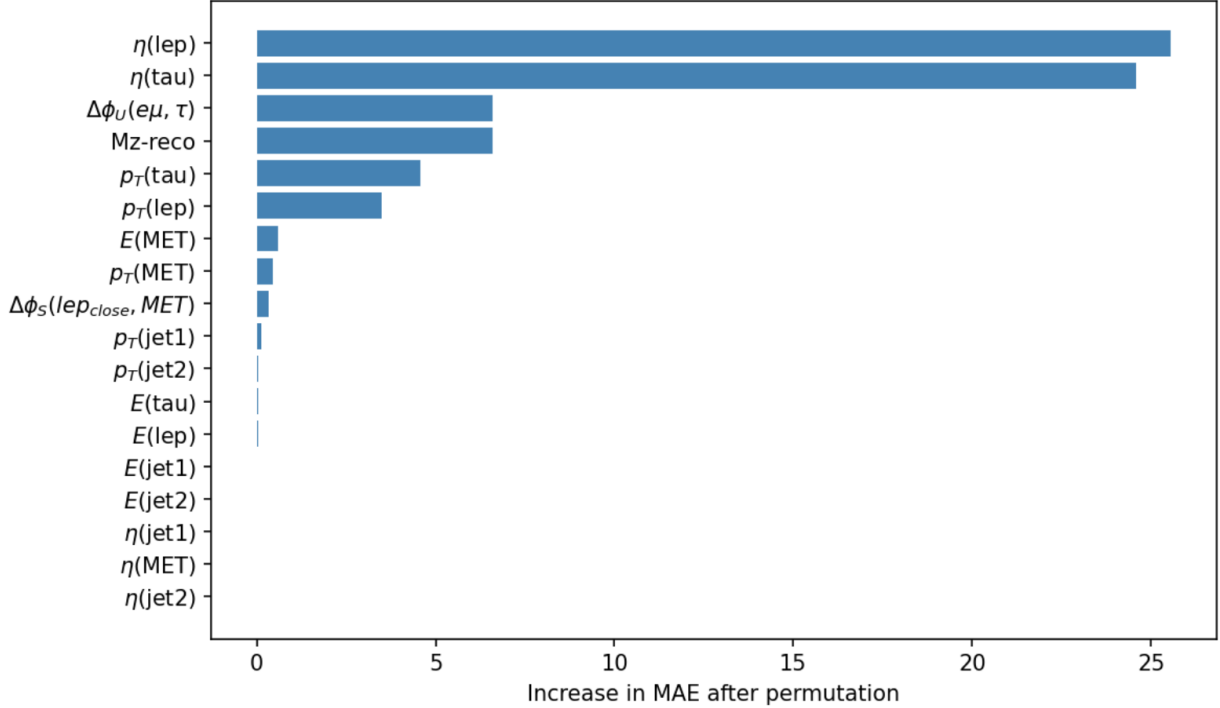


Figure 12: Feature importances for the DNN on Drell–Yan test data. The absolute increase in median absolute error is measured relative to the baseline of 2.10 GeV (Figure 10).

Transformer attention metrics provide additional insight. For the basic transformer, layer 0 ex-

hibits near-maximal entropy (1.6) and low head diversity (2.4%), suggesting that the layer is effectively ‘frozen’ and may act as a filter for subsequent layers. Layer 1 shows lower entropy and good diversity, indicating that attention is focused on fewer tokens, producing a richer representation. For the interaction transformer, entropy decreases between the two layers, implying information compression, while both layers maintain significant diversity. The higher diversity in Layer 2 likely contributes to its superior performance relative to the basic transformer.

Metric	Transformer	Inter-Transformer
Ln(Num. Input Tokens)	1.6	2.7
Layer 0 Entropy	1.6	2.5
Layer 1 Entropy	0.73	1.6
Layer 0 Diversity	2.4%	19%
Layer 1 Diversity	28%	68%

Table 3: Entropy and Diversity metrics for each of the layers in the basic and interaction transformer trained on Drell–Yan data. The entropy is bounded between 0 and the natural log of the number of input tokens.

5.3 VBF Models on Drell–Yan Data and Vice-Versa

To assess the generalization capabilities of the trained models, each one was evaluated on the dataset opposite to the one it was trained on. As shown on the left of Figure 13, models trained on VBF data and applied to the Drell–Yan dataset tend to underestimate the Z -peak. Nevertheless, the overall mass distribution shape is largely preserved, indicating that these models have learned generalizable mass-reconstruction skills rather than overfitting to a single event type, albeit with a systematic shift.

On the right of Figure 13, models trained on Drell–Yan data and applied to the VBF dataset tend to overestimate the peak and lose the peak shape, particularly for the DNN and basic transformer models. This suggests that the Drell–Yan-trained models do not generalize as well to other event types, likely due to overtraining on the large Drell–Yan dataset.

Interestingly, the interaction transformer performs best in this cross-validation test. This is likely because it has access to the largest pool of inputs and includes pair-wise particle information, allowing it to learn more general relationships rather than focusing on dataset-specific kinematics.

It is important to note that this represents a low-level test of generalization. As discussed in Appendix A, certain variables, such as momentum, exhibit dataset-dependent biases. These shifts likely contribute to the residual peak misalignment in the left panel of Figure 13. A more robust cross-validation could be achieved by removing extreme outliers from both datasets, which distort feature scaling, thereby providing a clearer assessment of true generalization capability.

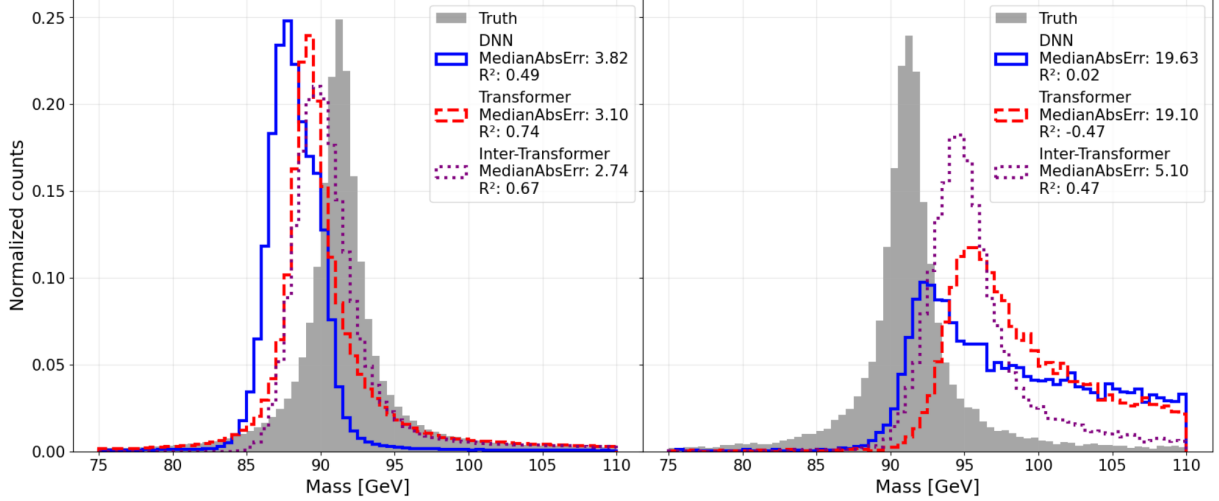


Figure 13: Plots of the performance of models trained on VBF data and evaluated on Drell–Yan test data (left). And performance of models trained on Drell–Yan data and evaluated on VBF test data (right). The region 75–110 GeV is displayed with MAE and R^2 scores for each model.

Notably, testing showed that only a few epochs of fine-tuning, starting from the pre-trained weights of one model, were sufficient for it to adapt and perform well on the other dataset.

6 Discussion

Both DNN and transformer models achieve good accuracy when the input variables are properly scaled and the architecture is appropriately tuned. This tuning process was more challenging for the VBF dataset due to its smaller size, making model convergence slower and more sensitive to hyperparameters. In contrast, the larger Drell–Yan dataset allowed all models to converge more reliably and achieve improved performance.

Hand-tuning was generally easier for DNNs because feature importance plots directly indicate which inputs influence the predictions. Transformers are more difficult to interpret, as their attention mechanisms provide only indirect insights into how the model prioritizes information. Across both datasets, the right tail of the mass distribution is consistently reconstructed better than the left tail (Figures 8 and 11). This is likely due to kinematic and training factors: low-mass events often have a larger fraction of energy carried by neutrinos, leaving less information in the visible decay products. These events are also sparsely populated in the training data, so the model naturally prioritizes the dense Z -peak region. Additionally, low- p_T particles in the left tail result in higher relative MET uncertainties, further challenging reconstruction.

All models display horizontal and vertical features in the 2D correlation plots, indicating regions where predictions for similar events diverge or where the model guesses the Z -peak despite different true masses. Further investigation could involve examining input distributions for events in these regions to understand why both DNNs and transformers struggle.

The results suggest that model complexity and dataset size interact in nontrivial ways. For the VBF dataset, the basic transformer outperformed the interaction transformer, showing that overly complex architectures can underperform on small datasets. For the Drell–Yan dataset, all models improved significantly, with transformers fitting the Z -peak well; however, it remains difficult to determine from the 2D plots which model best avoids vertical features due to the density of events.

Transformer attention metrics provide partial insight into model behavior. As seen in Tables 2 and 3, high entropy with low diversity in the first layer may act as a filter to enhance the second layer, but it is unclear whether this represents a meaningful physical understanding or merely a numerical artifact. Additional metrics and analyses are needed to more clearly identify which features are driving the final prediction.

This study demonstrates that particle transformers can be used for di-tau mass reconstruction and can slightly outperform DNNs. However, the complexity of transformers makes them more sensitive to small datasets, which may limit their practical utility. Preliminary tests of model generalization, such as applying VBF-trained models to Drell–Yan data (Figure 13), suggest some transferability, though further work is needed, including careful handling of extreme events and post-scaling shifts (Appendix A). Transfer learning from pre-trained weights shows promise, with rapid convergence observed within 10–20 epochs, but requires further investigation.

Future improvements could include more sophisticated sampling techniques, such as inverse KDE, to better cover sparsely populated regions of phase space. Additionally, systematic studies of regions in the 2D histograms where models struggle, coupled with analysis of the corresponding input distributions, could provide valuable insights for further optimizing both DNN and transformer architectures.

7 Conclusion

Overall, this work demonstrates that both Deep Neural Networks (DNNs) and Transformers are capable of reconstructing the mass of di-tau events, provided careful hyperparameter tuning and appropriate input data scaling.

Training and tuning were more straightforward on the larger Drell–Yan dataset with 6.2 million events, although training times ranged from 20 minutes to 2 hours. In contrast, the smaller VBF dataset with 195 thousand events allowed for faster training (2–10 minutes) but limited the models’ ability to learn, as evidenced by the closer agreement between predicted and true masses in the Drell–Yan case (Figures 10 vs. 7). In both datasets, all ML models significantly outperformed the traditional collinear approximation and Missing Mass Calculator, showing improved correlation with the true Z -boson mass (Figures 6, 8, and 11). Notably, the basic transformer performed best on the VBF data and the interaction transformer performed best on the Drell–Yan data.

Feature importance analyses for the DNNs indicate consistent patterns across datasets: particle-level observables such as the lepton’s η and p_T are highly valued, while the human-engineered

collinear mass (M_Z -reco) retains moderate importance.

The inner workings of the Transformer models were more difficult to interpret. Metrics such as attention entropy and head diversity provided only a superficial understanding of model behavior. Future studies could benefit from more sophisticated visualization techniques and analyses of input patterns in specific regions of the mass correlation plots, which may help explain persistent vertical and horizontal features observed across all models.

Finally, there is preliminary evidence that models trained on VBF data can generalize to Drell–Yan events (Figure 13), although differences in variable scaling and extreme outliers limit this performance. Future work could explore additional channels with similar final states and remove extreme events to improve consistency between datasets after scaling, further enhancing model generalization.

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8 Appendix

A Scaling and Distribution of Training Variables

In machine learning, input variables must be properly scaled to prevent features with disproportionately large magnitudes from dominating the model’s learning process. Depending on the

physical characteristics of each variable, different scaling strategies may be applied. For example, variables with long tails or outliers are often log-scaled, while quantities that are strictly positive may be normalized to the range $[0, 1]$.

A common qualitative check for the effectiveness of scaling is the ability of a variable to separate output classes. In classification tasks such as jet tagging, an ideally scaled variable would exhibit clear separation between classes. However, in our case, the task is regression, where the target is continuous. This makes such separation less meaningful, as there is effectively an infinite number of "classes," and the distinction between them is inherently blurred.

Table 4 summarizes the three common scaling methods used in this work, along with their mathematical definitions. Figure 14 illustrates the effect of scaling on a subset of input variables for the VBF and Drell–Yan datasets. To avoid data leakage, it is crucial to compute statistics such as the mean and standard deviation or the minimum and maximum (for min-max scaling) exclusively on the training dataset. These values are then applied to scale the validation and test datasets to prevent information leakage from unseen data into the training process.

As seen in Figure 14, the kinematic distributions of electrons, muons, and taus are similar between the VBF and Drell–Yan datasets. However, the p_T distributions appear shifted. This is a consequence of the difference in dataset sizes: the larger Drell–Yan dataset contains a higher probability of extremely high- p_T events, which skews the min-max scaling, effectively shifting the distribution to the left. Such differences complicate the generalization of a VBF-trained model to Drell–Yan data. One possible mitigation strategy is to remove extreme outliers that disproportionately affect the scaling, thereby improving consistency between datasets.

Closed-form formulas		Short descriptions	
Scaling	Formula	Input Var	Scaling
LogMinMax	$x' = \frac{\log(x) - \log(x_{\min})}{\log(x_{\max}) - \log(x_{\min})}$	Energy	LogMinMax
Standard	$x' = \frac{x - \mu}{\sigma}$	Eta	Standard
Cos/Sin	$x' = (\cos \phi, \sin \phi)$	Phi	Cos/Sin
		p_T	LogMinMax

Table 4: Scaling formulas used for the ML inputs. Closed mathematical formulas for the scaling on the left while example of scalers chosen on the right.

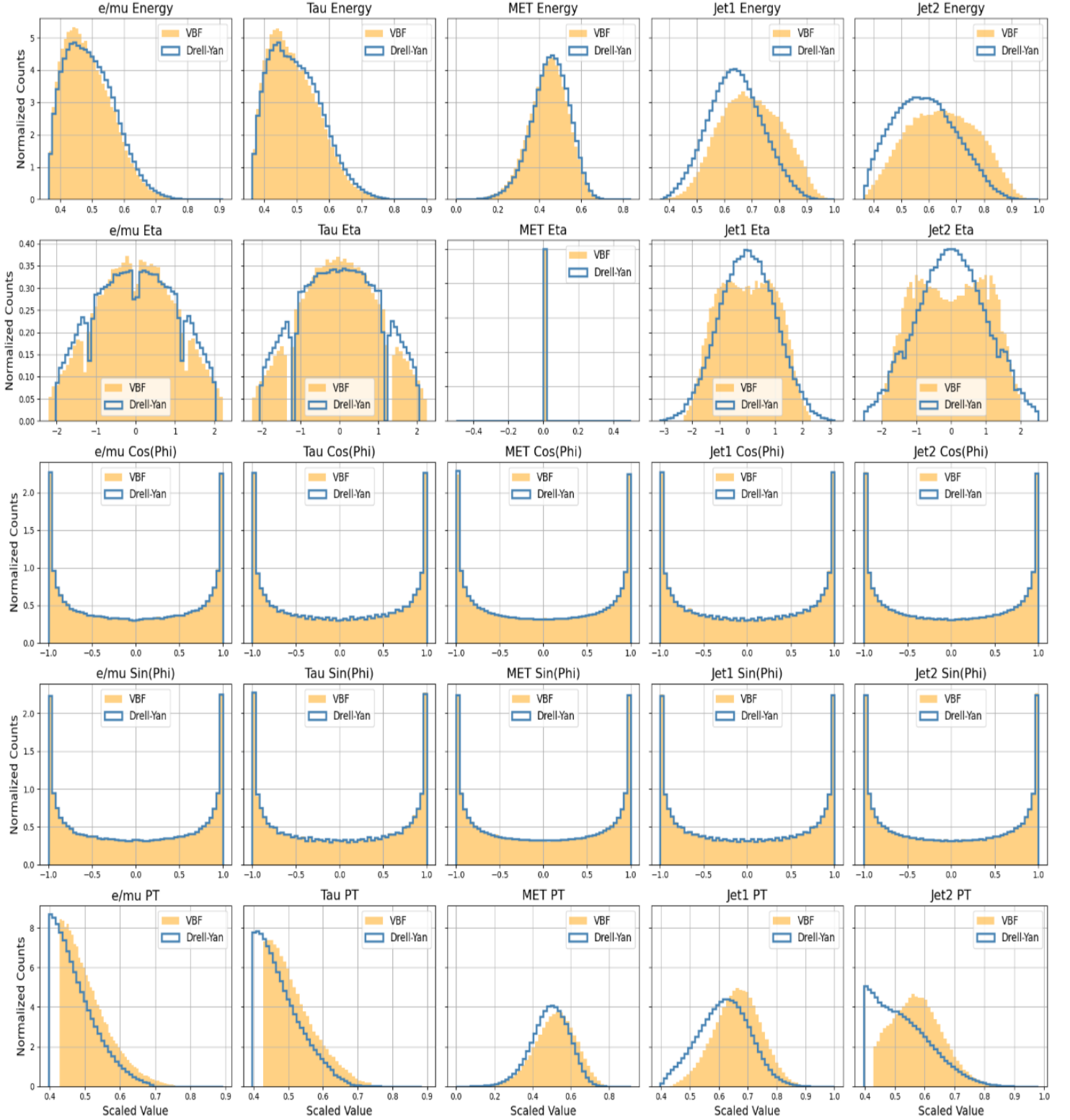


Figure 14: A subset of the scaled input data for the VBF and Drell-Yan samples. Each row represents a different feature: energy, eta, cos(phi), sin(phi) and p_T . Each column represents a different particle: e/mu, tau, MET, jet-1 and jet-2.